

The UrtheCast SAR-XL Multi-Band, Multi-Aperture Spaceborne SAR System

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Abstract— UrtheCast is currently developing the SAR-XL technology for its OptiSAR™ Constellation that comprises 8 tandem pairs of SAR and Optical Satellites in two orbit planes. This paper briefly outlines the OptiSAR™ mission and then describes the SAR-XL instrument that is carried by the SAR satellite. This novel technology is the world's first dual-frequency, circular/quad polarization, multi-aperture, fully-active phased-array and digitally-beamformed (in both azimuth and elevation) space-based Synthetic Aperture Radar (SAR). This instrument is highly flexible and can be used to generate many different types of beam modes enabling a large number of applications. The baseline beam modes used for the OptiSAR™ mission are described and their performance capabilities are given. Additional advanced beam modes that are possible using the SAR-XL technology are also identified. Finally, the modular nature of the instrument is also described that enables the SAR-XL to be easily configured for different antenna sizes with up to 16 apertures that can enable a totally new class of performance.

Keywords—SAR, Spaceborne, digital beamforming, multi-aperture, phased-array, OptiSAR™

I. INTRODUCTION

This paper describes the new patent pending SAR-XL technology that is used in UrtheCast's OptiSAR™ Constellation. A brief description of the OptiSAR™ mission is provided followed by a description of the SAR-XL instrument and associated products.

II. OPTISAR™ CONSTELLATION

The OptiSAR™ Constellation comprises 8 tandem pairs of SAR and Optical Satellites divided into two orbit planes. The 4 tandem pairs will be equi-spaced around an orbit plane, where each tandem pair consists of a leading SAR satellite, which uses UrtheCast's SAR-XL technology for its payload, and a trailing Optical satellite that is following approximately 2 minutes behind the SAR satellite. Figure 1 shows the OptiSAR™ tandem satellite pair configuration.

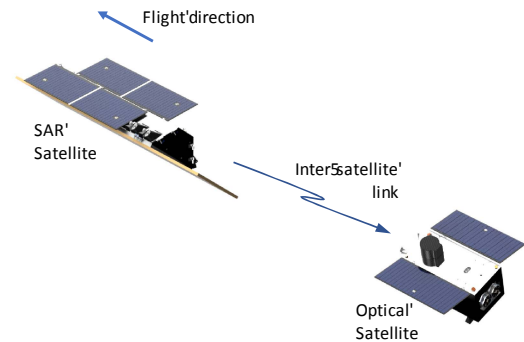


Figure 1: UrtheCast OptiSAR™ Constellation Tandem Satellite Pair

The first orbit plane is a Sun-Synchronous Orbit (SSO) with a 10:30 a.m. equator crossing time that is a commonly used orbit for Earth Observation missions, and the second plane is a Mid-Inclination Orbit (MIO) with a ~45 deg inclination. The MIO is used to provide ultra-high revisit in mid latitude regions of the earth where the bulk of the world's population reside. Both orbit planes have a satellite altitude of 450 km. The orbit plane configuration and the satellite phasing within each orbit plane is given in Figure 2.

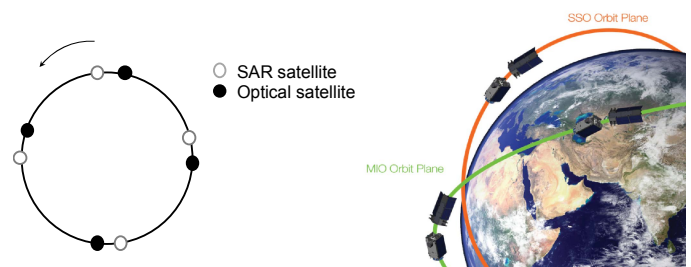


Figure 2: Arrangement of SAR-Optical Pairs in Each Orbital Plane

The OptiSAR™ constellation represents a capability shift in space-based remote sensing. The optical satellite uses a high-resolution (0.5m GSD) dual-mode (multi-spectral pushbroom and RGB colour video) optical camera, while the SAR satellite uses UrtheCast's SAR-XL technology that provides a dual-band (X-band and L-band) SAR instrument configured with 6 fully independent apertures. The SAR-XL payload provides 1-metre class X-band (VV polarization) imagery and 5-meter class fully polarimetric L-band imagery.

A key feature of the OptiSAR™ constellation is the cross-cueing capability between the SAR and optical satellites which enables the system to have a wide swath surveillance mode with the SAR satellite and, using onboard processing, detect and classify objects of interest (e.g., ships), and then automatically task the optical satellite to take a high-resolution image or video of the selected objects. The cross-cueing concept of operation is further described in Figure 3.

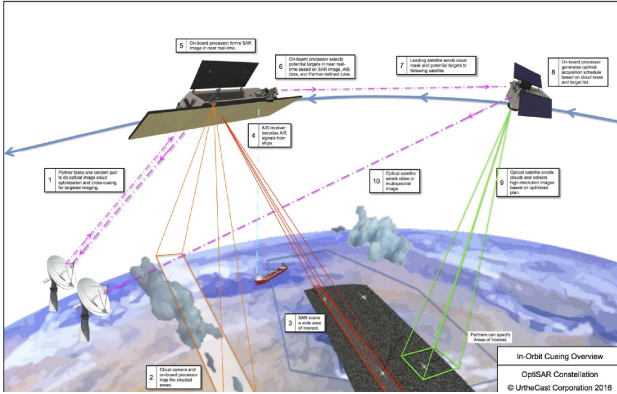


Figure 3: UrtheCast OptiSAR™ Cross-Cueing ConOps

III. SAR-XL INSTRUMENT DESCRIPTION

UrtheCast's SAR-XL technology is the world's first dual-frequency, quad polarization, multi-aperture, fully-active phased-array and all digitally-beamformed (in both azimuth and elevation) space-based Synthetic Aperture Radar (SAR). The SAR-XL instrument is architected and designed to be highly flexible in terms of operational beam modes and also modular in nature so the antenna can be sized to achieve the performance required. Each aperture is essentially a complete SAR instrument unto itself, so that the SAR antenna can be made up of a selected number of individual apertures. The flexibility provided by the instrument can serve a wide variety of applications that includes commercial geanalytics applications such as crop monitoring, forest structure modelling and biomass estimation, soil moisture retrieval, as well as very high resolution surveillance and wide-swath maritime domain surveillance for ship detection, ice monitoring and ocean winds [1,2].

The SAR-XL instrument has a dual-frequency (L & X band) shared aperture phased-array antenna, driven by multiple digital receiver-exciter. Each antenna aperture segment is 1.8 m (in elevation) x 1.0 m (in azimuth). The OptiSAR™ constellation SAR Satellite uses an antenna with 6 dual-frequency apertures, although the SAR-XL design can accommodate 16 apertures. These apertures are combined digitally to form the transmit beam, and to form multiple simultaneous receive beams, in both azimuth and elevation.

This architecture, as well as other features of the multiple digital receiver exciters, results in a Multi-Aperture Multi-Frequency Digitally Beamformed SAR with full Multi-Input-Multi-Output (MIMO) capabilities [3,4,5]. To further extend the dimensionality of the trade space that this MIMO SAR architecture enables, SAR-XL takes the MIMO architecture further, providing digital simultaneous beamforming of

multiple azimuth and elevation beams, at both X-Band and L-Band, from the same aperture. It is these capabilities that allow a relatively small antenna to meet the requirement of wide swath surveillance modes without excessive range ambiguities.

Each of the dual-frequency apertures has an independent synchronized receiver-exciter requiring only a reference oscillator, a clock, and a synchronization pulse provided to it externally. This allows easily combining any number of apertures to form the full antenna which provides a level of modularity that has not been demonstrated in a space-borne SAR in the past. The independent synchronized receiver-exciter are implemented with versatile parameterized state machines embedded in state-of-the-art FPGAs, providing further opportunities for agile reconfiguration and implementation of advanced SAR modes. Each aperture is independently capable of orchestrating complex modes that include the baseline modes of SpotLight, StripMap and ScanSAR, as well as advanced SAR operational modes, such as the Very High Resolution Multi-Aperture StripMap Mode at X-Band, producing 1m class resolution in continuous StripMap.

On the transmit side, the transmit pulse agility and flexibility supports both linear and non-linear FM pulses, allowing improved range impulse response without loss in NESZ. Additionally, the SAR-XL supports single, dual, slant and circular transmit polarization options. On the receive side, the within-PRI agility of the digital beamforming is capable of supporting modes such as Scan on Receive (SCORE), while the multiple simultaneous digital beams and dual frequencies can support modes such as the Multiple Simultaneous Beam Dual Frequency ScanSAR used for the wide swath surveillance. Additionally, the SAR-XL supports linear and circular receive polarization options.

Each aperture is capable of independent transmit pulses and elevation beams, as well as independent azimuth and elevation receive beams, receive centre-frequencies and bandwidths. More importantly, there is no RF harnessing between these independent synchronized panels. The panels are interconnected with high-speed digital interfaces, allowing an unprecedented flexibility in multi-channel azimuth digital beamforming (using multiple azimuth beams) in support of Multi-Aperture Processing Scheme (MAPS).

The Transmit Receive Modules (TRMs) are unconventional in that they do not have any phase shifters; with all the phase shifting and phase corrections being done digitally within the receiver-exciter. Implemented with Gallium Nitride, the TRMs deviate from conventional TRMs in producing pulse powers that are an order of magnitude larger than is considered normal for a TRM. These digital phase shifters enable formation of multiple simultaneous beams in receive, in both elevation and azimuth, thus can support SCORE (elevation), moving target velocity determination (azimuth), as well as several novel experimental modes currently being researched.

Combining MAPS and SCORE allows for significant increase of swath while maintaining resolution and image quality [3,4].

The SAR-XL dual-frequency shared aperture antenna uses highly efficient radiators integrated directly into the structural elements in such a manner that every piece of metal forms a dual function of electromagnetic radiation, power distribution, and structural strength. The antenna itself is also the support structure for the electronics units. In this manner the SAR-XL sensor can operate within the mass and size constraints imposed by the nature of a commercial mission.

Testing of the SAR-XL antenna in an anechoic chamber has confirmed radiation efficiencies (excluding corporate feed losses) and cross-polarization purity on each aperture on the order of 70% or better and 40dB (L-band) respectively. These levels of performance were achieved by using a patent pending technology that integrates all of the electromagnetic and structural elements into a single antenna panel.

IV. SAR-XL BEAMS AND MODES SPECIFICATIONS

The UrtheCast OptiSAR™ constellation will fly a 6 aperture configuration SAR-XL resulting in an antenna 6m in length (azimuth) by 1.8m (elevation). This configuration of the SAR-XL instrument has been modeled and analyzed in detail, and the models were confirmed by ground tests of the prototype antenna. The baseline beam modes are illustrated in Figure 4 with a brief description provided below.

SpotLight Mode products will have a nominal scene size of 10km (cross-track) by 5km (along-track) with an X-band ground resolution as fine as 1.0m by 1.1m (Range x Azimuth), typical ambiguity ratios of -25dB and typical NESZ of between -20dB and -24dB.

Multi-Aperture StripMap Mode products will have a nominal swath width of 7.5km with an X-band ground resolution as fine as 1.0m by 1.0m (Range x Azimuth), typical ambiguity ratios of -19dB and typical NESZ of between -19dB and -23dB. The multi-aperture stripmap mode products have approximately the same resolution as spotlight mode products, but they are capable of yielding long strips providing much greater coverage.

Dual-Band StripMap Mode products will have a nominal swath width of 10km. The X-band ground resolution will be 1.0m by 3.5m (Range x Azimuth), with typical range ambiguity ratios of -20dB to -25dB, typical azimuth ambiguity ratios of -19dB to -22 dB, and typical NESZ of between -23dB and -26dB. For single, dual and compact polarizations, the L-band ground resolution will be 7.5m x 7.5m, with typical ambiguity ratios of -22dB and typical NESZ of between -23dB to -26dB. For L-band quad polarization, the swath width will be 7.5 km and the ground resolution will be 16m x 7.5m (Range x Azimuth), with a typical range ambiguity ratio of -26dB, typical azimuth ambiguity ratio of -21dB, and a typical NESZ of between -28dB and -30dB.

ScanSAR Mode products will have a nominal swath width of 140km, with an L-band ground-range resolution of 30m by 30m (Range x Azimuth), typical range ambiguity ratio between -21dB to -24dB, typical azimuth ambiguity ratio between -19dB to -22dB, and typical NESZ of between -23dB to -26dB.

V. SAR-XL – AN EXTENDABLE ARCHITECTURE

The modular nature of the SAR-XL architecture is such that it can easily be extended up to as many as a 16 aperture configuration potentially resulting in an antenna 16m in length (azimuth) by 1.8m (elevation) which can be used for a future mission. It is also possible to go for even larger number of apertures with modest modifications to the Antenna Interface and Control Unit (AICU). This provides unprecedented flexibility in the antenna configuration that is possible using the same basic modular design elements which can significantly reduce the non-recurring engineering costs when using this technology for other missions.

Using more apertures would provide a significant improvement in the performance of the system, which includes finer resolution, increased swath width, lower ambiguity ratios and improved NESZ. For example, the L-band could potentially provide a 150 km width swath and a 2.7 m x 2.8 m resolution.

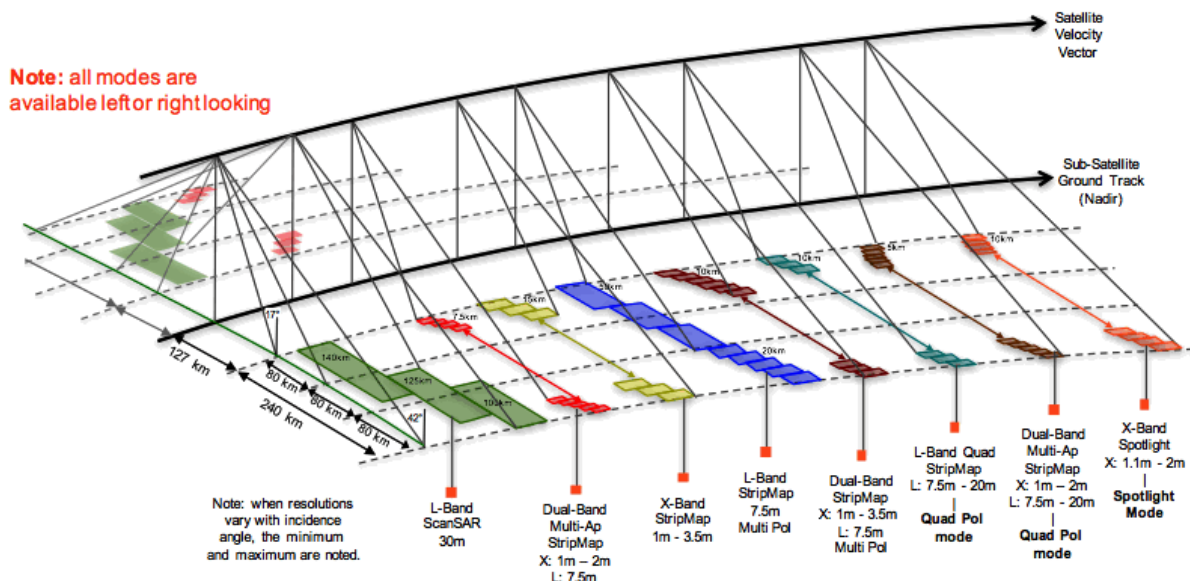


Figure 4: Baseline UrtheCast OptiSAR™ SAR-XL Beams and Modes

VI. CONCLUSIONS

The patent pending SAR-XL technology and associated mission that is being developed for the UrtheCast OptiSAR™ constellation is truly novel, establishing several world first achievements in significant aspects such as dual-frequency, multi-aperture and fully digital beamforming in both azimuth and elevation. The performance of this technology has been demonstrated by ground tests of prototype hardware which have confirmed the performance models. The modular nature of the SAR-XL technology, where each dual-frequency aperture is a complete SAR instrument unto itself and with no need for complex RF interfaces between panels, is also unprecedented. This offers tremendous flexibility in the configuration of the antenna to meet the desired performance levels for the mission while minimizing non-recurring engineering costs.

The shift in imaging capability realized through this SAR-XL technology has only begun with the suite of baseline products described herein; yet there remains many other capabilities yet to be explored further toward breaking through on the fundamental challenges of high-resolution and wide swath imaging.

REFERENCES

- [1] K. Beckett, "UrtheCast Second-Generation Earth Observation Sensors," Proceedings of the 36th International Symposium on Remote Sensing of Environment, 2015, Berlin, Germany, pp. 1069–1073.
- [2] K. Ouchi, "Recent Trend and Advance of Synthetic Aperture Radar with Selected Topics," Remote Sensing, 2013, pp. 716–807.
- [3] M. Younis, F. Queiroz de Almeida, P. López-Dekker and G. Krieger, "Techniques and Modes for Multi-Channel SAR Instruments," Proceedings of the 11th European Conference on Synthetic Aperture Radar, 2016, Hamburg, Germany, pp. 812–817, 2016.
- [4] G. Adamiuk, C. Heer and M. Ludwig, "DBF Technology Development for Next Generation of ESA C-Band SAR Mission," Proceedings of the 11th European Conference on Synthetic Aperture Radar, 2016, Hamburg, Germany, pp. 825–828.
- [5] G. Krieger, N. Gebert, M. Younis and A. Moreira, "Advanced Synthetic Aperture Radar Based on Digital Beamforming and Waveform Diversity," Proceedings of the 2008 IEEE Radar Conference, 2008, Rome, Italy, pp. 767–722.
- [6] F. Queiroz de Almeida and G. Krieger, "Multichannel Staggered SAR Azimuth Sample Regularization," Proceedings of the 11th European Conference on Synthetic Aperture Radar, 2016, Hamburg, Germany, pp. 1067–1072.
- [7] G. Krieger, N. Gebert, M. Younis, F. Bordoni, A. Patyuchenko and A. Moreira, "Advanced Concepts for Ultra-Wide-Swath SAR Imaging," Proceedings of the 7th European Conference on Synthetic Aperture Radar, 2008, Friedrichshafen, Germany, pp. 1–4.